Precision Placement in Scaled Dot-Product Attention

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Abstract. We study scaled dot-product attention as a precision-placement problem. Rather than asking whether one global low-precision dtype is acceptable, we ask which individual stages of the forward pass can safely run in low precision: storage of the query, key, and value tensors, the logit product QK^T, logit storage, the rowwise softmax, and the final value product. The present draft still analyzes an analytically transparent sketch-based baseline, but its main purpose is to motivate a broader decomposition of attention error into logit perturbation, softmax perturbation, and final output perturbation. The accompanying code now includes exact-attention precision sweeps with policy-level controls and diagnostics for relative logit error, rowwise softmax error, and output error.

Keywords: mixed-precision arithmetic, attention, transformers, numerical stability, floating-point arithmetic

1 Introduction

Scaled dot-product attention [1] is the core computational primitive of transformer architectures. As models grow, the memory and compute cost of the attention forward pass motivates the use of reduced-precision arithmetic. The natural question is a *precision-placement* question: which stages of the forward pass can safely run in low precision, and which still require higher-precision accumulation or reduction?

The relevant stages are: storage of the query, key, and value tensors; the logit product QK^T; logit storage; the rowwise softmax; and the final value product. A global dtype comparison—the usual approach—conflates these stages and cannot identify which *locations* are numerically sensitive. This report instead decomposes the attention pipeline and asks, for each stage separately, whether low precision there produces acceptable error.

The analytical core is Proposition 2, a local perturbation bound for exact attention that separates the effect of logit perturbations (weighted by $\|V\|_2$) from value perturbations (weighted by $\sqrt{n}$ from the row-stochastic spectral norm). This decomposition directly predicts which precision locations are sensitive: logit errors propagate through softmax with unit Lipschitz constant, while value errors propagate with a $\sqrt{n}$ factor that is typically moderate. The sharpest failure mode is therefore low-precision softmax, which can amplify tiny logit perturbations into large probability shifts even when the logit error itself is small.

Sketching-based attention approximations [2] [3] [4] [5] [6] are retained as a methodological baseline: they provide a controlled stress test in which the error decomposition can be verified under a known approximation floor. But they are not the main contribution.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both finite-dimensional Hilbert spaces endowed with their standard Euclidean inner products. We view matrices in $\mathbb{R}^{n\times d}$ as Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \to \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\mathrm{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact scaled logit matrix, viewed as an operator on $\mathcal{H}_n$, is

$$\mathcal{L}(Q,K) := d^{-1/2} QK^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q,K,V) := \Sigma(\mathcal{L}(Q,K))V.$$

3 Analytical Core

The analysis begins with a stability property of the rowwise softmax map, which is used in both the exact-attention and sketching bounds.

Proposition 1 (Rowwise softmax is Hilbert–Schmidt Lipschitz)**.** Let $L, M \in \mathbb{R}^{n\times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\mathrm{HS}} \le \|L - M\|_{\mathrm{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n\times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\mathrm{HS}} \le \|L - M\|_{\mathrm{HS}}\, \|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \to \mathbb{R}^n$ be the softmax map on vectors with Jacobian $D\sigma(x) = \mathrm{Diag}(p) - pp^T$ where $p = \sigma(x)$. For any $z \in \mathbb{R}^n$, $z^T D\sigma(x)\, z = \sum_i p_i z_i^2 - (\sum_i p_i z_i)^2 \le \|z\|_2^2$, so $\|D\sigma(x)\|_2 \le 1$ for all x. The mean value theorem gives $\|\sigma(x) - \sigma(y)\|_2 \le \|x - y\|_2$; applying this row by row and squaring yields the HS bound. The second assertion follows from $\|XV\|_{\mathrm{HS}} \le \|X\|_{\mathrm{HS}}\|V\|_2$. □

3.1 Exact-Attention Perturbation Bound

The next proposition is the main analytical result. It decomposes the attention error into a logit-driven term and a value-driven term, directly predicting the relative sensitivity of different precision placements.

Proposition 2 (Exact-attention local perturbation bound)**.** Let

$$A = \Sigma(L)V, \qquad \widehat{A} = \Sigma(\widehat{L})\widehat{V},$$

where $L, \widehat{L} \in \mathbb{R}^{n\times n}$ and $V, \widehat{V} \in \mathbb{R}^{n\times d}$. Writing

$$\Delta L := \widehat{L} - L, \qquad \Delta V := \widehat{V} - V,$$

one has

$$\|A - \widehat{A}\|_{\mathrm{HS}} \le \|\Delta L\|_{\mathrm{HS}}\,\|V\|_2 + \|\Sigma(\widehat{L})\|_2\,\|\Delta V\|_{\mathrm{HS}}.$$

In particular, since $\Sigma(\widehat{L})$ is row-stochastic and therefore $\|\Sigma(\widehat{L})\|_2 \le \sqrt{n}$,

$$\|A - \widehat{A}\|_{\mathrm{HS}} \le \|\Delta L\|_{\mathrm{HS}}\,\|V\|_2 + \sqrt{n}\,\|\Delta V\|_{\mathrm{HS}}.$$

Proof. Add and subtract $\Sigma(\widehat{L})V$ and apply the triangle inequality:

$$\|A - \widehat{A}\|_{\mathrm{HS}} \le \|\Sigma(L)V - \Sigma(\widehat{L})V\|_{\mathrm{HS}} + \|\Sigma(\widehat{L})(V - \widehat{V})\|_{\mathrm{HS}}.$$

The first term is bounded by Proposition 1:

$$\|\Sigma(L)V - \Sigma(\widehat{L})V\|_{\mathrm{HS}} \le \|L - \widehat{L}\|_{\mathrm{HS}}\,\|V\|_2.$$

For the second,

$$\|\Sigma(\widehat{L})(V - \widehat{V})\|_{\mathrm{HS}} \le \|\Sigma(\widehat{L})\|_2\,\|\Delta V\|_{\mathrm{HS}}.$$

Combining these gives the stated bound. □

Remark 3 (Interpretation for precision placement)**.** Proposition 2 directly predicts the relative danger of different precision choices:

(1) **Logit perturbations** ($\Delta L \neq 0$, $\Delta V = 0$): The error scales as $\|\Delta L\|_{\mathrm{HS}}\|V\|_2$. Since the softmax is 1-Lipschitz (Proposition 1), logit perturbations are *not amplified* by softmax itself. However, when logit perturbations are amplified into probability errors, the amplification ratio ρ_{softmax} can still be large on inputs where the softmax is nearly one-hot—a regime the experiments expose.

(2) **Value perturbations** ($\Delta L = 0$, $\Delta V \neq 0$): The error scales as $\sqrt{n}\,\|\Delta V\|_{\mathrm{HS}}$. This $\sqrt{n}$ factor is moderate for typical sequence lengths but can become significant for very long sequences.

(3) **Low-precision softmax**: When Σ itself is computed in low precision, the Lipschitz bound from Proposition 1 no longer applies, and the effective perturbation of the probability matrix can be much larger than $\|\Delta L\|_{\mathrm{HS}}$. The experiments confirm this: ρ_{softmax} spikes to $\sim 10^3$ on transformer activations when softmax is run in bf16/fp16, while remaining < 1 when only storage is low precision.

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3.2 Supporting Background: Sketching as a Controlled Baseline

The following propositions analyze a sketched surrogate $\widetilde{\mathcal{A}}_S(Q,K,V) = \Sigma(d^{-1/2}QSS^*K^*)V$ obtained by replacing the full logit product with a low-dimensional projection through $S \in \mathbb{R}^{d\times s}$ with $s < d$. These results are retained as a methodological baseline: the sketching error provides a known approximation floor against which the finite-precision error can be isolated and studied.

Proposition 4 (Deterministic sketching bound)**.** Let $U \subseteq \mathcal{H}_d$ be the joint row space of Q and K, equivalently the subspace spanned by the columns of $[Q^T \;\; K^T]$, and let $r := \dim U$. Assume that S is an ϵ-subspace embedding for U, meaning

$$\left|\langle S^*u, S^*v\rangle - \langle u,v\rangle\right| \le \epsilon \,\|u\|_2 \,\|v\|_2 \qquad \text{for all } u,v \in U.$$

Equivalently, if $U_0 \in \mathbb{R}^{d\times r}$ is any orthonormal basis matrix for U, then $\|U_0^T SS^*U_0 - I_r\|_2 \le \epsilon$. Then

$$\|\mathcal{L}(Q,K) - \widetilde{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \le \frac{\epsilon}{\sqrt{d}}\,\|Q\|_{\mathrm{HS}}\,\|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q,K,V) - \widetilde{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \le \frac{\epsilon}{\sqrt{d}}\,\|Q\|_{\mathrm{HS}}\,\|K\|_2\,\|V\|_2.$$

Proof. Let $U_0 \in \mathbb{R}^{d\times r}$ have orthonormal columns spanning the joint row space of Q and K. Write $Q = Q_U U_0^T$ and $K = K_U U_0^T$, where $Q_U = QU_0 \in \mathbb{R}^{n\times r}$ and $K_U = KU_0 \in \mathbb{R}^{n\times r}$. Then

$$QK^* - QSS^*K^* = Q_U(I_r - U_0^T SS^*U_0)K_U^T.$$

The subspace-embedding condition is exactly $\|U_0^T SS^*U_0 - I_r\|_2 \le \epsilon$, so

$$\|QK^* - QSS^*K^*\|_{\mathrm{HS}} \le \|Q_U\|_{\mathrm{HS}}\,\|I_r - U_0^T SS^*U_0\|_2\,\|K_U^T\|_2 \le \epsilon\,\|Q\|_{\mathrm{HS}}\,\|K\|_2,$$

where we used $\|Q_U\|_{\mathrm{HS}} = \|Q\|_{\mathrm{HS}}$ and $\|K_U\|_2 = \|K\|_2$. Multiplying by $d^{-1/2}$ gives the logit bound. The attention-output bound follows from Proposition 1. □

Proposition 5 (Probabilistic subspace embedding guarantee)**.** Let $U \subseteq \mathcal{H}_d$ be an r-dimensional subspace. Let $S \in \mathbb{R}^{d\times s}$ have i.i.d. entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For $\epsilon \in (0,1)$ and $\delta \in (0,1)$, if

$$s \;\ge\; C\,\frac{r + \ln(1/\delta)}{\epsilon^2}\,,$$

for a universal constant C, then with probability at least $1-\delta$, S is an ϵ-subspace embedding for U.

Proof. Let $U_0 \in \mathbb{R}^{d\times r}$ be a matrix whose columns form an orthonormal basis for U. Then S is an ϵ-subspace embedding for U if and only if $\|U_0^T SS^*U_0 - I_r\|_2 \le \epsilon$. The matrix $U_0^T SS^*U_0 = (S^*U_0)^*(S^*U_0)$ has the same distribution as G^*G/s where $G \in \mathbb{R}^{s\times r}$ has i.i.d. standard normal entries. Standard JL concentration [2] gives

$\Pr(\|G^*G/s - I_r\|_2 > \epsilon) \le 2\exp(-c\,s\epsilon^2 + r)$ for a universal constant $c > 0$, provided s is large enough relative to r/ϵ^2. The stated condition with an appropriate C ensures the failure probability is at most δ. □

Proposition 6 (Finite-precision error for sketched logits)**.** Let $\widehat{\mathcal{L}}_S(Q,K)$ denote the logit matrix computed in floating-point arithmetic with unit roundoff u, and let $\gamma_s := su/(1-su)$ for $su < 1$. Then

$$\|\widetilde{\mathcal{L}}_S(Q,K) - \widehat{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \le \frac{\gamma_s\,\|QS\|_{\mathrm{HS}}\,\|KS\|_{\mathrm{HS}}}{\sqrt{d}}\,.$$

Proof. Write the sketched logit product as $d^{-1/2}AB^*$ with $A = QS \in \mathbb{R}^{n\times s}$ and $B = KS \in \mathbb{R}^{n\times s}$. The standard forward-error bound for matrix multiplication [7] gives

$$|\widehat{AB}^* - AB^*| \le \gamma_s\,|A|\,|B^*|,$$

entrywise in absolute value. Taking the Frobenius (Hilbert–Schmidt) norm and applying the Cauchy–Schwarz inequality for matrices,

$$\|\widehat{AB}^* - AB^*\|_{\mathrm{HS}} \le \gamma_s\,\||A|\,|B^*|\|_{\mathrm{HS}} \le \gamma_s\,\|A\|_{\mathrm{HS}}\,\|B\|_{\mathrm{HS}}.$$

Multiplying by $d^{-1/2}$ yields the bound. □

Remark 7 (Scope of the finite-precision bound)**.** The bound in Proposition 6 models the error in the final sketched logit product $(QS)(KS)^*$, assuming that QS and KS are given exactly. The experiments measure a broader implementation error that also includes storage quantization and low-precision projection effects. ⌟

Combining the subspace embedding guarantee with the finite-precision error yields the sketching main theorem.

Theorem 8 (Mixed-precision sketched attention)**.** Under the hypotheses of Propositions 5 and 6, with probability at least $1-\delta$,

$$\|\mathcal{A}(Q,K,V) - \widehat{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \le \frac{1}{\sqrt{d}}\Big(\epsilon\,\|Q\|_{\mathrm{HS}}\,\|K\|_2 + \gamma_s\,\|QS\|_{\mathrm{HS}}\,\|KS\|_{\mathrm{HS}}\Big)\|V\|_2.$$

The first term is the sketching error; the second is the finite-precision error.

Proof. By Proposition 1 and the triangle inequality,

$$\|\mathcal{A} - \widehat{\mathcal{A}}_S\|_{\mathrm{HS}} \le (\|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\mathrm{HS}} + \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}})\,\|V\|_2.$$

Apply Proposition 4 to the first term and Proposition 6 to the second. □

Remark 9 (Design rule from the sketching theorem)**.** Theorem 8 yields a concrete joint design rule. For float16 ($u_{16} \approx 5\times 10^{-4}$) and typical sketch dimensions $s \le 64$, we have $s\cdot u_{16} \le 0.032$, which is smaller than typical sketch tolerances $\epsilon \ge 0.1$. For float32 ($u_{32} \approx 6\times 10^{-8}$), the condition is essentially always satisfied. For float8 E4M3 ($u_{\mathrm{fp8}} \approx 0.0625$), even $s = 16$ gives $s\cdot u_{\mathrm{fp8}} \approx 1.0$, which is comparable to ϵ—so the finite-precision term is no longer clearly lower order.

4 Numerical Experiments

4.1 Exact Attention Under Precision Policies

For each precision policy, we evaluate the exact attention pipeline

$$Q, K, V \to L = QK^*/\sqrt{d} \to P = \Sigma(L) \to A = PV$$

against a `float64` reference. The primary diagnostics are the relative logit error

$$E_L = \frac{\|L - \widehat{L}\|_{\mathrm{HS}}}{\|L\|_{\mathrm{HS}}},$$

the mean rowwise softmax error

$$E_P = \frac{1}{n}\sum_{i=1}^{n} \|p_i - \widehat{p}_i\|_1,$$

the relative output error

$$E_A = \frac{\|A - \widehat{A}\|_{\mathrm{HS}}}{\|A\|_{\mathrm{HS}}},$$

and the softmax amplification ratio

$$\rho_{\mathrm{softmax}} = \frac{\frac{1}{n}\sum_{i=1}^{n} \|p_i - \widehat{p}_i\|_1}{\frac{1}{n}\sum_{i=1}^{n} \|\ell_i - \widehat{\ell}_i\|_\infty}.$$

We test the following policy family: `fp32_reference`, `bf16_safe`, `fp16_safe`, low-precision logit storage, low-precision softmax, and low-precision value accumulation. The "safe" policies keep storage low precision but preserve fp32 accumulation and fp32 softmax. This separation is the core experimental point: the question is not whether one global dtype works, but which *locations* are sensitive.

Policy	E_L	E_P	E_A	ρ_{softmax}
`fp32_reference`	8.2e-8	8.5e-8	1.4e-7	0.41
`bf16_safe`	2.4e-3	1.9e-3	2.8e-3	0.35
`fp16_safe`	2.9e-4	2.1e-4	3.2e-4	0.32
`bf16_low_logit`	2.9e-3	2.3e-3	3.4e-3	0.34
`fp16_low_logit`	3.5e-4	2.8e-4	4.4e-4	0.33
`bf16_low_softmax`	2.9e-3	2.7e-3	3.7e-3	0.41
`fp16_low_softmax`	3.5e-4	3.3e-4	4.7e-4	0.39
`bf16_low_value`	2.4e-3	1.9e-3	3.5e-3	0.35
`fp16_low_value`	2.9e-4	2.1e-4	4.4e-4	0.32

Table 10 / Exact-attention errors on synthetic Gaussian data ($n \in \{16, 32\}$, $d = 16$, averaged over 2 seeds). "Safe" policies use low-precision storage with fp32 accumulation and softmax.

Tables 10 and 11 report representative policy-level numbers from Gaussian and transformer-activation sweeps respectively. On Gaussian data (Table 10), all policies remain low-error

Policy	E_L	E_P	E_A	ρ_{softmax}
`fp32_reference`	1.8e-8	3.0e-8	5.0e-8	4.6e4
`bf16_safe`	6.1e-4	3.0e-8	1.8e-3	1.9
`fp16_safe`	3.3e-4	3.0e-8	3.5e-4	3.4
`bf16_low_logit`	3.9e-4	3.0e-8	1.8e-3	2.8
`fp16_low_logit`	3.9e-4	3.0e-8	3.5e-4	2.8
`bf16_low_softmax`	3.9e-4	1.2e-5	1.8e-3	1.1e3
`fp16_low_softmax`	3.9e-4	1.2e-5	3.4e-4	1.1e3
`bf16_low_value`	6.1e-4	3.0e-8	1.8e-3	1.9
`fp16_low_value`	3.3e-4	3.0e-8	3.4e-4	3.4

Table 11 / Exact-attention errors on a single attention head extracted from a pretrained transformer (`sshleifer/tiny-distilbert-base-cased`, layer 0, head 0; $n = 16$, $d = 1$). The softmax amplification ratio ρ_{softmax} spikes under low-precision softmax, confirming the prediction from Proposition 2 and Remark 3.

because the softmax is not near-saturated: $\rho_{\text{softmax}} < 1$ uniformly, meaning the softmax actually *damps* logit perturbations. The ordering $E_L < E_P < E_A$ is consistent across all policies, and the differences between "safe" and "low" variants are modest.

The transformer activations (Table 11) reveal a qualitatively different regime. The baseline ρ_{softmax} for `fp32_reference` is $\sim 10^4$ because the denominator (rowwise logit perturbation) is at machine epsilon while the numerator is even smaller; this large ratio is a numerical artifact of dividing by a near-zero baseline. The important comparison is between "safe" policies ($\rho \sim 1$–3) and "low_softmax" policies ($\rho \sim 10^3$). Low-precision softmax amplifies the tiny logit perturbation ($E_L \approx 4 \times 10^{-4}$) into a softmax error E_P that is two orders of magnitude larger relative to the safe baseline. This is exactly the failure mode predicted by Remark 3(3): when softmax itself is computed in low precision, the Lipschitz guarantee from Proposition 1 no longer holds.

Figure 14 summarizes these policy-level results together with the residual-depth experiment discussed below. The most important empirical message is that low-precision storage of Q, K, V is usually much safer than low-precision softmax or low-precision logit storage. This is precisely the kind of location-specific statement that a global dtype comparison would miss.

4.2 Residual Depth Propagation

To test how local precision errors accumulate across depth, we evaluate an attention-only residual stack of the form

$$x_{\ell+1} = x_\ell + hF_\ell(x_\ell)$$

with self-attention blocks F_ℓ. The metric of interest is the relative terminal-state error

$$E_{\text{depth}} = \frac{\|x_L - \widehat{x}_L\|_{\text{HS}}}{\|x_L\|_{\text{HS}}}.$$

Table 12 reports E_{depth} for both Gaussian and transformer-activation residual stacks. On Gaussian data, error grows modestly with depth for all policies: the fp16-safe error

	Gaussian		Transformer	
Policy	depth 4	depth 8	depth 2	depth 4
fp32_reference	7.2e-8	8.8e-8	5.2e-8	2.2e-8
bf16_safe	3.3e-3	4.7e-3	3.4e-3	1.1e-3
fp16_safe	4.5e-4	6.7e-4	1.6e-4	1.7e-4
bf16_low_logit	3.3e-3	4.9e-3	3.4e-3	1.1e-3
fp16_low_logit	4.5e-4	6.3e-4	1.6e-4	1.7e-4
bf16_low_softmax	3.6e-3	4.7e-3	3.4e-3	1.1e-3
fp16_low_softmax	4.3e-4	6.8e-4	1.6e-4	1.7e-4
bf16_low_value	3.6e-3	5.3e-3	3.1e-3	1.2e-3
fp16_low_value	5.3e-4	7.8e-4	6.6e-4	6.1e-4

Table 12 / Relative terminal-state error E_{depth} for attention-only residual stacks (Gaussian data, $n = d = 16$, averaged over 2 seeds). Depth $\ell = 4$ uses $h = 0.25$; depth $\ell = 8$ uses $h = 0.125$.

roughly doubles from depth 4 to depth 8, while the bf16-safe error grows by roughly 50%. The low-precision value policies show the steepest depth growth, consistent with the $\sqrt{n}\|\Delta V\|_{\mathrm{HS}}$ term in Proposition 2 accumulating across layers. On transformer activations, the residual scaling $h = 1/\text{depth}$ keeps errors approximately constant or even decreasing with depth, reflecting the stabilizing effect of small residual contributions.

4.3 A Stress-Test Baseline

The experiments in this subsection probe the decomposition of Theorem 8 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

(1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);

(2) $\widetilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;

(3) $\widehat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching error is $E_{\mathrm{sk}} := \|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\mathrm{HS}}$ and the finite-precision error is $E_{\mathrm{fp}} := \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}}$. The total error is $E_{\mathrm{tot}} := \|\mathcal{L} - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}}$. The same decomposition is recorded for the attention output. We test four precision configurations: `fp32` (all stages in float32), `full_fp16` (storage, projection, final logit product, and softmax in float16), `mixed_fp16` (storage and projection in float16, followed by a float32 final logit product and float32 softmax), and `fp8_store` (storage and projection in float8 E4M3, followed by a float32 final logit product and float32 softmax).

The sketch $S \in \mathbb{R}^{d\times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries; the scaling is chosen so that $\mathbb{E}[SS^*] = I_d$. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds. Data and sketch seeds are independent. The full numerical results are available in the committed CSV file `results/attention_sweep.csv`.

Since the synthetic Q and K are i.i.d. Gaussian, their joint row space is typically full-dimensional, so for $d = 64$ and $s \leq 32$ the subspace-embedding regime from Proposition 5 is not expected to hold. The experiment should therefore be read as an intentionally under-sketched stress test of the error decomposition, not as evidence that such small feature sketches preserve attention accurately.

Table 13 reports absolute HS errors of the logits for $n = 256$, $d = 64$, together with the ratio $E_{\text{sk}}/E_{\text{fp}}$; the output errors showed the same qualitative ordering and are omitted.

Config	s	E_{sk}	E_{fp}	E_{tot}	$E_{\text{sk}}/E_{\text{fp}}$
`fp32`	8	7.22e2	1.2e-4	7.22e2	5.9e6
	16	5.03e2	1.1e-4	5.03e2	4.4e6
	24	4.27e2	9.4e-5	4.27e2	4.5e6
	32	3.70e2	9.1e-5	3.70e2	4.1e6
`full_fp16`	8	7.22e2	4.1e-1	7.22e2	1.8e3
	16	5.03e2	2.9e-1	5.03e2	1.8e3
	24	4.27e2	2.6e-1	4.27e2	1.7e3
	32	3.70e2	2.3e-1	3.70e2	1.6e3
`mixed_fp16`	8	7.22e2	3.8e-1	7.22e2	1.9e3
	16	5.03e2	2.6e-1	5.03e2	1.9e3
	24	4.27e2	2.4e-1	4.27e2	1.8e3
	32	3.70e2	2.1e-1	3.70e2	1.8e3
`fp8_store`	8	7.22e2	5.0e1	7.21e2	1.4e1
	16	5.03e2	3.4e1	5.01e2	1.5e1
	24	4.27e2	3.0e1	4.26e2	1.4e1
	32	3.70e2	2.6e1	3.70e2	1.4e1

Table 13 / Hilbert–Schmidt errors of the logit matrix ($n = 256$, $d = 64$, averaged over 3 seeds). $E_{\text{sk}} = \|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{fp}} = \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{tot}} = \|\mathcal{L} - \widehat{\mathcal{L}}_S\|_{\text{HS}}$; $\|\mathcal{L}\|_{\text{HS}} \approx 256$.

Three observations stand out. First, for fp32 and fp16, $E_{\text{sk}}/E_{\text{fp}}$ is consistently large (10^3–10^6), confirming that the dominant error source is the randomized sketch. Second, `full_fp16` and `mixed_fp16` have finite-precision errors of the same order, consistent with Proposition 6: storage quantization dominates any benefit from higher-precision accumulation. Third, fp8 storage substantially increases the finite-precision error (from $\sim$0.3 to $\sim$30 in absolute terms), but the sketching error still dominates by a factor of $\sim$14–15. The results support the error decomposition and illustrate the degradation predicted by Theorem 8 for increasing $s \cdot u$, but they do not demonstrate a high-quality efficient-attention approximation or a full inversion of the error ratio.

5 Outlook

The main contribution of this report is the precision-placement decomposition of Proposition 2 and its empirical validation: low-precision softmax is the clearest failure mode in exact attention, while low-precision storage of Q, K, V with fp32 accumulation is typically safe. The sketching baseline (Theorem 8) confirms that the finite-precision term remains lower order as long as $s \cdot u \ll \epsilon$.

Several directions remain open. First, the current experiments use small problem sizes ($n \leq 256$, $d \leq 64$); scaling to realistic transformer dimensions will require either GPU acceleration or the JAX backend scaffold already present in the repository. Second,

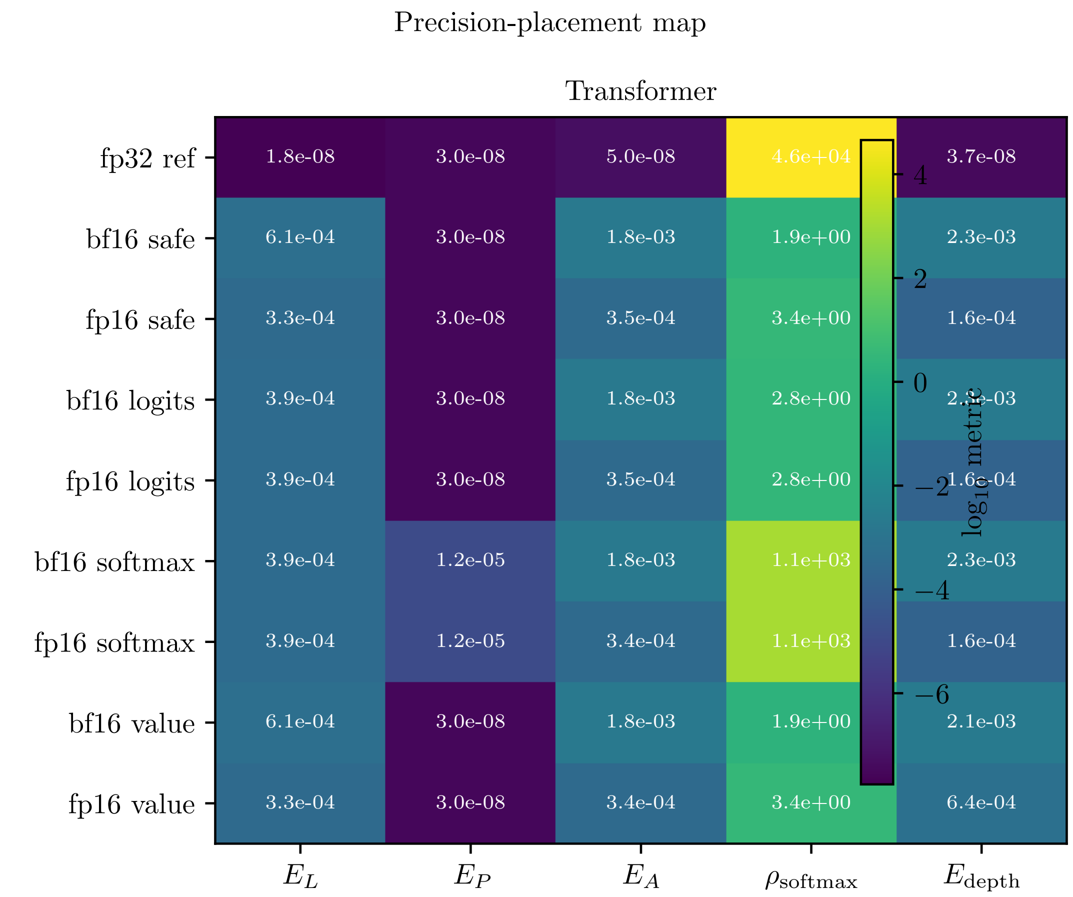

Figure 14 / Precision-placement map for exact attention. Rows correspond to explicit precision policies, columns to the principal diagnostics E_L, E_P, E_A, ρ_{softmax}, and E_{depth}. The Gaussian panel gives the synthetic baseline; the transformer panel uses a fixed head from a pretrained model. Safe policies with fp32 accumulation and fp32 softmax remain low-error, whereas low-precision softmax is the clearest failure mode through its sharp increase in ρ_{softmax}.

FP8 policies are reported only under an explicit quantization model; a more complete treatment would include hardware-specific scaling and dequantization. Third, the phase-diagram analogy from Decelle et al. [8] suggests mapping not only average error but also regime changes: when does low-precision logit storage remain harmless, when does low-precision softmax begin to amplify, and when does no practical low-precision policy preserve the reference behavior? The current data already hints at these transitions but does not yet chart them systematically. Fourth, the JAX-focused mixed-precision framework of Graefe and Trimpe [9] provides a natural implementation path for the safe/unsafe policy distinctions identified here.

On the analytical side, the $\sqrt{n}$ bound on $\|\Sigma(\widehat{L})\|_2$ is likely loose for near-doubly-stochastic attention matrices; a refined bound that exploits the concentration of softmax rows could tighten the value-perturbation term. The logit-perturbation term could similarly be refined by replacing the worst-case $\|V\|_2$ with an average-case or data-dependent quantity.

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